

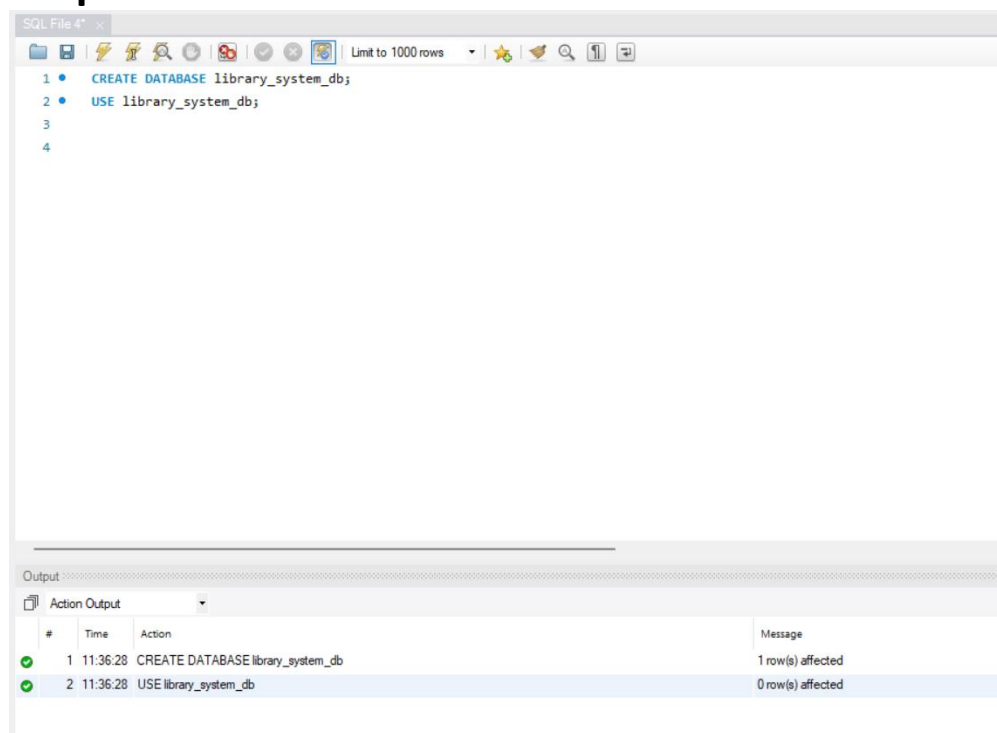
Week - 1

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Question :

A local library is currently managing its books and members using a physical notebook, which has resulted in several operational problems such as misplaced books, difficulty tracking borrowed items, and inaccurate member records. To modernize its operations, the library plans to develop a database system called **BookFlow**. As a database developer, your task is to create the SQL foundation for this system. First, create a new database named **bookflow_db**. Then, design two tables: a **Books** table containing the columns **book_id**, **title**, **isbn**, and **published_year**, and a **Members** table containing the columns **member_id**, **full_name**, and **email**. Ensure data integrity by setting **book_id** and **member_id** as **PRIMARY KEYs**, applying a **NOT NULL** constraint to the **title** column so that every book must have a name, enforcing **UNIQUE** constraints on both **isbn** and **email** to prevent duplicate entries, and adding a **CHECK** constraint to ensure that the **published_year** is not in the future (for example, it must be less than 2027). Finally, insert three sample records into both the **Books** and **Members** tables to verify that the database structure and constraints are functioning correctly. This exercise will help you develop skills in schema definition, data integrity enforcement, and the use of SQL Data Definition Language (DDL) commands such as **CREATE DATABASE**, **CREATE TABLE**, and **INSERT** statements.

Step 1 — Create Database



The screenshot shows a SQL IDE window titled "SQL File 4" with a toolbar and a code editor. The code editor contains the following SQL statements:

```
1 • CREATE DATABASE library_system_db;
2 • USE library_system_db;
3
4
```

Below the code editor is an "Output" panel with a dropdown menu set to "Action Output". It displays a table with the following data:

#	Time	Action	Message
1	11:36:28	CREATE DATABASE library_system_db	1 row(s) affected
2	11:36:28	USE library_system_db	0 row(s) affected

Step 2 — Create Books Table (DDL + Constraints)

```
Week-1* x
Limit to 1000 rows
1  -- Step 1: Create Database
2  • CREATE DATABASE library_system_db;
3  • USE library_system_db;
4  -- Step 2: Create Books Table
5  • CREATE TABLE Books(
6      book_id INT PRIMARY KEY,
7      title VARCHAR(100) NOT NULL,
8      isbn VARCHAR(20) UNIQUE,
9      published_year INT CHECK(published_year < 2027)
10 );
```

Output

Action Output

#	Time	Action	Message
✓ 1	19:08:11	CREATE DATABASE library_system_db	1 row(s) affected
✓ 2	19:08:15	USE library_system_db	0 row(s) affected
✓ 3	19:08:48	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(...	0 row(s) affected

```
Week-1* x
Limit to 1000 rows
1  -- Step 1: Create Database
2  • CREATE DATABASE library_system_db;
3  • USE library_system_db;
4  -- Step 2: Create Books Table
5  • CREATE TABLE Books(
6      book_id INT PRIMARY KEY,
7      title VARCHAR(100) NOT NULL,
8      isbn VARCHAR(20) UNIQUE,
9      published_year INT CHECK(published_year < 2027)
10 );
11 -- Step 3: Create Members Table
12 • CREATE TABLE Members(
13     member_id INT PRIMARY KEY,
14     full_name VARCHAR(100) NOT NULL,
15     email VARCHAR(100) UNIQUE
16 );
```

Output

Action Output

#	Time	Action	Message
✓ 1	19:08:11	CREATE DATABASE library_system_db	1 row(s) affected
✓ 2	19:08:15	USE library_system_db	0 row(s) affected
✓ 3	19:08:48	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(...	0 row(s) affected
✓ 4	19:09:33	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, email ...	0 row(s) affected

Week-1*

```

11 -- Step 3: Create Members Table
12 • CREATE TABLE Members(
13     member_id INT PRIMARY KEY,
14     full_name VARCHAR(100) NOT NULL,
15     email VARCHAR(100) UNIQUE
16 );
17 -- Step 4: Describe Books Table
18 • DESCRIBE Books;

```

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	
title	varchar(100)	NO		NULL	
isbn	varchar(20)	YES	UNI	NULL	
published_year	int	YES		NULL	

Result 1

Output

Action Output

#	Time	Action	Message
✓ 1	19:08:11	CREATE DATABASE library_system_db	1 row(s) affected
✓ 2	19:08:15	USE library_system_db	0 row(s) affected
✓ 3	19:08:48	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(...	0 row(s) affected
✓ 4	19:09:33	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, email...	0 row(s) affected
✓ 5	19:10:02	DESCRIBE Books	4 row(s) returned

Week-1

```

13     member_id INT PRIMARY KEY,
14     full_name VARCHAR(100) NOT NULL,
15     email VARCHAR(100) UNIQUE
16 );
17 -- Step 4: Describe Books Table
18 • DESCRIBE Books;
19 -- Step 5: Describe Members Table
20 • DESCRIBE Members;

```

Result Grid

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	
full_name	varchar(100)	NO		NULL	
email	varchar(100)	YES	UNI	NULL	

Result 2

Output

Action Output

#	Time	Action	Message
✓ 1	19:08:11	CREATE DATABASE library_system_db	1 row(s) affected
✓ 2	19:08:15	USE library_system_db	0 row(s) affected
✓ 3	19:08:48	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(...	0 row(s) affected
✓ 4	19:09:33	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, email...	0 row(s) affected
✓ 5	19:10:02	DESCRIBE Books	4 row(s) returned
✓ 6	19:10:39	DESCRIBE Members	3 row(s) returned

```
Week-1* x
Limit to 1000 rows

6      book_id INT PRIMARY KEY,
7      title VARCHAR(100) NOT NULL,
8      isbn VARCHAR(20) UNIQUE,
9      published_year INT CHECK(published_year < 2027)
10     );
11     -- Step 3: Create Members Table
12     CREATE TABLE Members(
13         member_id INT PRIMARY KEY,
14         full_name VARCHAR(100) NOT NULL,
15         email VARCHAR(100) UNIQUE
16     );
17     -- Step 4: Describe Books Table
18     DESCRIBE Books;
19     -- Step 5: Describe Members Table
20     DESCRIBE Members;
21     -- Step 6: Insert Records into Books Table
22     INSERT INTO Books VALUES
23     (1,'The Alchemist','9780061122415',1988),
24     (2,'Clean Code','9780132350884',2008),
25     (3,'Atomic Habits','9780735211292',2018);
```

Output

#	Time	Action	Message
2	19:08:15	USE library_system_db	0 row(s) affected
3	19:08:48	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR...	0 row(s) affected
4	19:09:33	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, e...	0 row(s) affected
5	19:10:02	DESCRIBE Books	4 row(s) returned
6	19:10:39	DESCRIBE Members	3 row(s) returned
7	19:11:13	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

```
Week-1* x
Limit to 1000 rows

20     DESCRIBE Members;
21     -- Step 6: Insert Records into Books Table
22     INSERT INTO Books VALUES
23     (1,'The Alchemist','9780061122415',1988),
24     (2,'Clean Code','9780132350884',2008),
25     (3,'Atomic Habits','9780735211292',2018);
26     -- Step 7: Display Books Table
27     SELECT * FROM Books;
```

Result Grid

	book_id	title	isbn	published_year
1	1	The Alchemist	9780061122415	1988
2	2	Clean Code	9780132350884	2008
3	3	Atomic Habits	9780735211292	2018
4		NULL	NULL	NULL

Books 3 x

Output

#	Time	Action	Message
3	19:08:48	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR...	0 row(s) affected
4	19:09:33	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, e...	0 row(s) affected
5	19:10:02	DESCRIBE Books	4 row(s) returned
6	19:10:39	DESCRIBE Members	3 row(s) returned
7	19:11:13	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
8	19:11:50	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned

Commands SQL File 4*

```

15     email VARCHAR(100) UNIQUE
16 );
17 -- Step 4: Describe Books Table
18 • DESCRIBE Books;
19 -- Step 5: Describe Members Table
20 • DESCRIBE Members;
21 -- Step 6: Insert Records into Books Table
22 • INSERT INTO Books VALUES
23 (1,'The Alchemist','9780061122415',1988),
24 (2,'Clean Code','9780132350884',2008),
25 (3,'Atomic Habits','9780735211292',2018);
26 -- Step 7: Display Books Table
27 • SELECT * FROM Books;
28 -- Step 8: Insert Records into Members Table
29 • INSERT INTO Members VALUES
30 (101,'Angela','angela@gmail.com'),
31 (102,'Smija','smija@gmail.com'),
32 (103,'Joel','joel@gmail.com');
33
34

```

Output

Action Output

#	Time	Action	Message
✓ 3	14:19:44	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, e...	0 row(s) affected
✓ 4	14:19:49	DESCRIBE Books	4 row(s) returned
✓ 5	14:19:56	DESCRIBE Members	3 row(s) returned
✓ 6	14:20:07	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	14:20:13	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
✓ 8	14:22:59	INSERT INTO Members VALUES (101,'Angela','angela@gmail.com'), (102,'Smija','smija@gmail.com'), (103,'Jo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Commands SQL File 4*

```

26 -- Step 7: Display Books Table
27 • SELECT * FROM Books;
28 -- Step 8: Insert Records into Members Table
29 • INSERT INTO Members VALUES
30 (101,'Angela','angela@gmail.com'),
31 (102,'Smija','smija@gmail.com'),
32 (103,'Joel','joel@gmail.com');
33 -- Step 9: Display Members Table
34 • SELECT * FROM Members;

```

Result Grid

member_id	full_name	email
101	Angela	angela@gmail.com
102	Smija	smija@gmail.com
103	Joel	joel@gmail.com
NULL	NULL	NULL

Members 4*

Output

Action Output

#	Time	Action	Message
✓ 4	14:19:49	DESCRIBE Books	4 row(s) returned
✓ 5	14:19:56	DESCRIBE Members	3 row(s) returned
✓ 6	14:20:07	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	14:20:13	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
✓ 8	14:22:59	INSERT INTO Members VALUES (101,'Angela','angela@gmail.com'), (102,'Smija','smija@gmail.com'), (103,'Jo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 9	14:23:41	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned


```
Commands SQL File 4* x
Limit to 1000 rows

20 • DESCRIBE Members;
21 -- Step 6: Insert Records into Books Table
22 • INSERT INTO Books VALUES
23 (1,'The Alchemist','9780061122415',1988),
24 (2,'Clean Code','9780132350884',2008),
25 (3,'Atomic Habits','9780735211292',2018);
26 -- Step 7: Display Books Table
27 • SELECT * FROM Books;
28 -- Step 8: Insert Records into Members Table
29 • INSERT INTO Members VALUES
30 (101,'Angela','angela@gmail.com'),
31 (102,'Smija','smija@gmail.com'),
32 (103,'Joel','joel@gmail.com');
33 -- Step 9: Display Members Table
34 • SELECT * FROM Members;
35 -- Step 10: Test Duplicate ISBN
36 • INSERT INTO Books VALUES
37 (4,'Fake Book','9780061122415',2000);
38
39
```

Output			
Action Output			
#	Time	Action	Message
✓ 5	14:19:56	DESCRIBE Members	3 row(s) returned
✓ 6	14:20:07	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	14:20:13	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
✓ 8	14:22:59	INSERT INTO Members VALUES (101,'Angela','angela@gmail.com'), (102,'Smija','smija@gmail.com'), (103,'Jo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 9	14:23:41	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
✗ 10	14:24:25	INSERT INTO Books VALUES (4,'Fake Book','9780061122415',2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'

```
Commands SQL File 4* x
Limit to 1000 rows

22 • INSERT INTO Books VALUES
23 (1,'The Alchemist','9780061122415',1988),
24 (2,'Clean Code','9780132350884',2008),
25 (3,'Atomic Habits','9780735211292',2018);
26 -- Step 7: Display Books Table
27 • SELECT * FROM Books;
28 -- Step 8: Insert Records into Members Table
29 • INSERT INTO Members VALUES
30 (101,'Angela','angela@gmail.com'),
31 (102,'Smija','smija@gmail.com'),
32 (103,'Joel','joel@gmail.com');
33 -- Step 9: Display Members Table
34 • SELECT * FROM Members;
35 -- Step 10: Test Duplicate ISBN
36 • INSERT INTO Books VALUES
37 (4,'Fake Book','9780061122415',2000);
38 -- Step 11: Test NULL Title
39 • INSERT INTO Books VALUES
40 (5,NULL,'999999999999',2010);
41
42
```

Output			
Action Output			
#	Time	Action	Message
✓ 6	14:20:07	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	14:20:13	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
✓ 8	14:22:59	INSERT INTO Members VALUES (101,'Angela','angela@gmail.com'), (102,'Smija','smija@gmail.com'), (103,'Jo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 9	14:23:41	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
✗ 10	14:24:25	INSERT INTO Books VALUES (4,'Fake Book','9780061122415',2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'
✗ 11	14:35:06	INSERT INTO Books VALUES (5,NULL,'999999999999',2010)	Error Code: 1048. Column 'title' cannot be null

Commands SQL File 4*

```

24 (2,'Clean Code','9780132350884',2008),
25 (3,'Atomic Habits','9780735211292',2018);
26 -- Step 7: Display Books Table
27 • SELECT * FROM Books;
28 -- Step 8: Insert Records into Members Table
29 • INSERT INTO Members VALUES
30 (101,'Angela','angela@gmail.com'),
31 (102,'Smija','smija@gmail.com'),
32 (103,'Joel','joel@gmail.com');
33 -- Step 9: Display Members Table
34 • SELECT * FROM Members;
35 -- Step 10: Test Duplicate ISBN
36 • INSERT INTO Books VALUES
37 (4,'Fake Book','9780061122415',2000);
38 -- Step 11: Test NULL Title
39 • INSERT INTO Books VALUES
40 (5,NULL,'9999999999999',2010);
41 -- Step 12: Test Future Publication Year
42 • INSERT INTO Books VALUES
43 (6,'Future Book','8888888888888',2030);
44

```

Output

Action Output

#	Time	Action	Message
7	14:20:13	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
8	14:22:59	INSERT INTO Members VALUES (101,'Angela','angela@gmail.com'), (102,'Smija','smija@gmail.com'), (103,'Jo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
9	14:23:41	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
10	14:24:25	INSERT INTO Books VALUES (4,'Fake Book','9780061122415',2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'
11	14:35:06	INSERT INTO Books VALUES (5,NULL,'9999999999999',2010)	Error Code: 1048. Column 'title' cannot be null
12	14:36:15	INSERT INTO Books VALUES (6,'Future Book','8888888888888',2030)	Error Code: 3819. Check constraint 'books_chk_1' is violated.

Commands SQL File 4*

```

29 • INSERT INTO Members VALUES
30 (101,'Angela','angela@gmail.com'),
31 (102,'Smija','smija@gmail.com'),
32 (103,'Joel','joel@gmail.com');
33 -- Step 9: Display Members Table
34 • SELECT * FROM Members;
35 -- Step 10: Test Duplicate ISBN
36 • INSERT INTO Books VALUES
37 (4,'Fake Book','9780061122415',2000);
38 -- Step 11: Test NULL Title
39 • INSERT INTO Books VALUES
40 (5,NULL,'9999999999999',2010);
41 -- Step 12: Test Future Publication Year
42 • INSERT INTO Books VALUES
43 (6,'Future Book','8888888888888',2030);
44 -- Step 13: Test Duplicate Email
45 • INSERT INTO Members VALUES
46 (104,'Angela Clone','angela@gmail.com');
47
48

```

Output

Action Output

#	Time	Action	Message
9	14:23:41	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
10	14:24:25	INSERT INTO Books VALUES (4,'Fake Book','9780061122415',2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'
11	14:35:06	INSERT INTO Books VALUES (5,NULL,'9999999999999',2010)	Error Code: 1048. Column 'title' cannot be null
12	14:36:15	INSERT INTO Books VALUES (6,'Future Book','8888888888888',2030)	Error Code: 3819. Check constraint 'books_chk_1' is violated.
13	14:36:59	INSERT INTO Members VALUES (104,'Harshika Clone','lharshika@gmail.com')	1 row(s) affected
14	14:37:29	INSERT INTO Members VALUES (104,'Angela Clone','angela@gmail.com')	Error Code: 1062. Duplicate entry '104' for key 'members.PRIMARY'